



1. **DESCRIPTION:** Participants will demonstrate their knowledge of plant life and general botany principles.
ATEAM OF UPTO: 2

APPROXIMATE TIME: 50 minutes

CALCULATOR: Class II

2. **EVENT PARAMETERS:**

- a. Each team may bring writing utensils.
- b. Each team may bring two Class II calculators.
- c. Each team may bring one (1) 8.5" x 11" sheet of paper that may contain information on both sides in any form and from any source. The sheet of paper may be laminated or placed in a sheet protector to increase durability. Affixed labels, as well as multiple sheets of paper (whether in a single sheet protector or not) are prohibited.
- d. Event Supervisors will provide live/preserved specimens, pictures, tables, graphs of data, microscopes, slides, and any other required equipment for the event. If used, toxic/irritating plants or specimens in liquid (e.g., Algae, protists) must be in closed, non-breakable containers.

3. **THE COMPETITION:**

- a. This event may be run as a paper pencil test or as a station test.
- b. Participants will be expected to understand the structure of plant cells, roots, stems, leaves, spore forming bodies and flowers, aspects of plant growth and differentiation, and the transport and storage of gases, water, and nutrition throughout the plant body.
- c. Participants should also have a broad knowledge of the major divisions between groups of plants (i.e., algae vs. multicellular plants, monocot vs. dicot, embryophytes vs. cryptogams, woody vs. herbaceous plants).
- d. In addition to the above listed topics, participants should know:
 - i. The history of botany
 - ii. Basic plant genetics and reproduction
 - iii. Photosynthesis
 - iv. Differences between the major taxonomic groups of plants
 - v. Paleo-botany and plant evolution
 - vi. The role of plants in global energy and nutrient cycles
 - vii. Use of plant materials by animals and humans
 - viii. Competition in the plant community
 - ix. Genetically Modified Organisms (GMOs)
 - x. Production of foodstuffs and plant products
 - xi. Plant diseases; including nutrient deficiencies and infections
 - xii. Principles of horticulture and aquaculture
 - xiii. Plant biochemistry
 - xiv. The roles of plants in medicine and environmental management
 - xv. Importance of plant diversity

4. **SCORING:**

- a. High Score wins.
- b. Preselected questions will be used to break ties.